Practical 10: AirPassengers Time Series Forecasting

Objective

The objective is to analyze the monthly international airline passengers dataset (`AirPassengers`) and forecast the passenger numbers for the next 12 months (the year 1961).

Dataset

- **Source**: Standard R dataset `AirPassengers`.
- **Observations**: 144 (Monthly data from 1949 to 1960).
- **Units**: 1,000s of passengers.

Methodology

1. **Visualization**: Plotted the original series to identify trend and seasonality.
2. **Decomposition**: Applied multiplicative decomposition because the seasonal variation increases as the trend increases.
3. **Stationarity Analysis**: Used KPSS test and ACF/PACF plots.
4. **Model Selection**: Used `auto.arima()` with a log transformation (`lambda=0`) to handle increasing variance.
5. **Validation**: Performed residual diagnostics (Ljung-Box test).
6. **Forecasting**: Projected values for the next 12 months.

Code Snippet

The analysis followed a structured approach, utilizing the `forecast` and `tseries` libraries:

1. Data Preparation

```
# Loading the built-in R AirPassengers dataset
data(AirPassengers)
ap <- AirPassengers
```

2. Exploratory Data Analysis & Decomposition

```
# Identifying trend and seasonality
plot(ap, main = "AirPassengers Time Series")
# Multiplicative decomposition because variance increases with level
ap_decomp <- decompose(ap, type = "multiplicative")
plot(ap_decomp)
```

3. Stationarity Analysis

```
# Testing for level stationarity
kpss.test(ap, null = "Level")
# ACF/PACF plots for seasonal patterns
acf(ap, lag.max = 48)
pacf(ap, lag.max = 48)
```

4. Model selection (with Log Transformation)

```
# Fit SARIMA with lambda=0 (Log-transformation) to stabilize variance
final_model <- auto.arima(ap, lambda = 0, seasonal = TRUE, stepwise =
FALSE, approximation = FALSE)
summary(final_model)
```

5. Goodness of Fit (Residual Diagnostics)

Before generating forecasts, we examine the residuals to ensure no patterns remain and the error term is white noise:

```
# Checkresiduals function calculates Ljung-Box test and plots ACF/Histogram
checkresiduals(final_model)
```

6. Forecasting as a Function

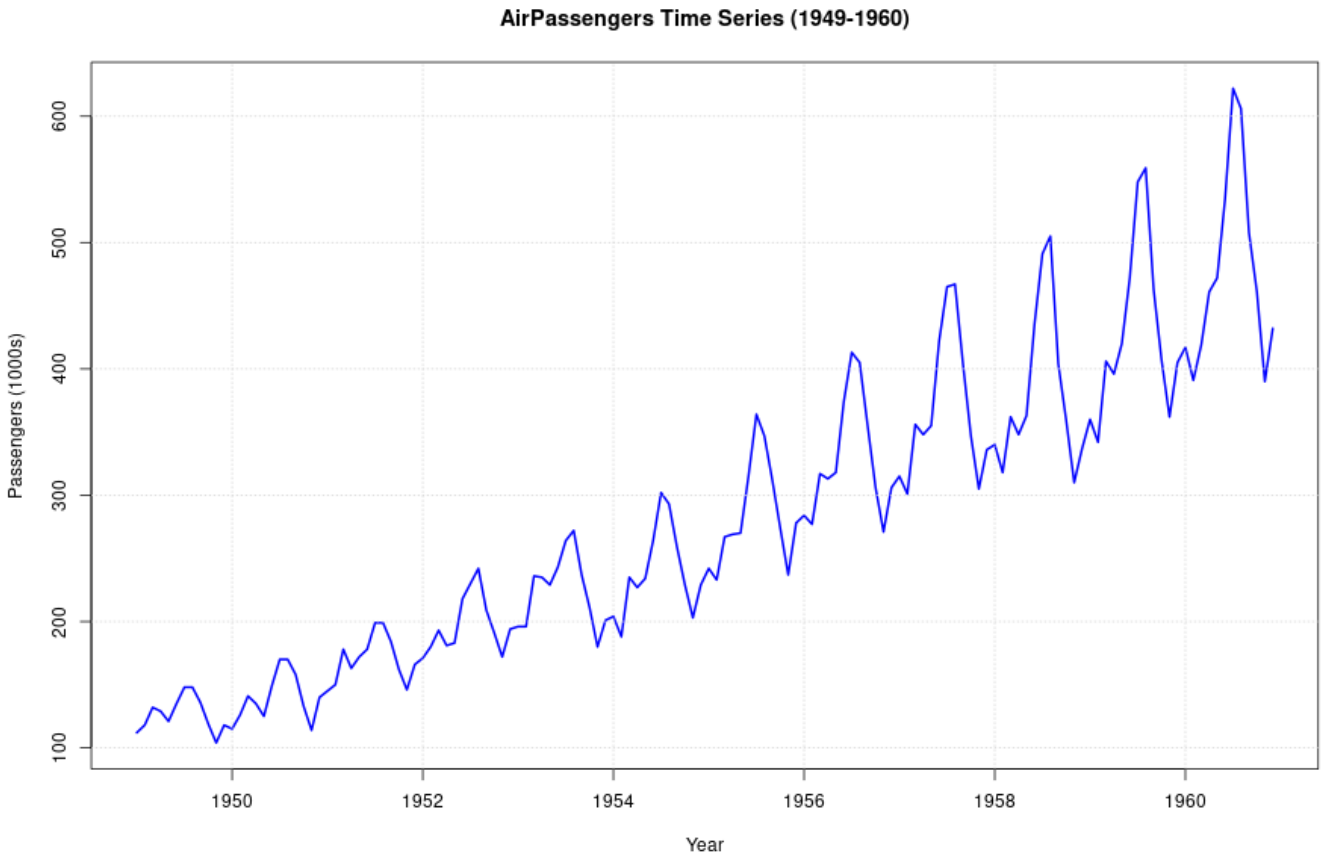
Once the model is validated, the forecasting function `forecast()` is applied. Since `lambda=0` was used, it automatically back-transforms the log-scale results back to original units:

```
# Forecasting the next 12 months (h=12)
ap_forecast <- forecast(final_model, h = 12)
# Plotting the forecast with seasonal behavior maintained
plot(ap_forecast)
```

Results

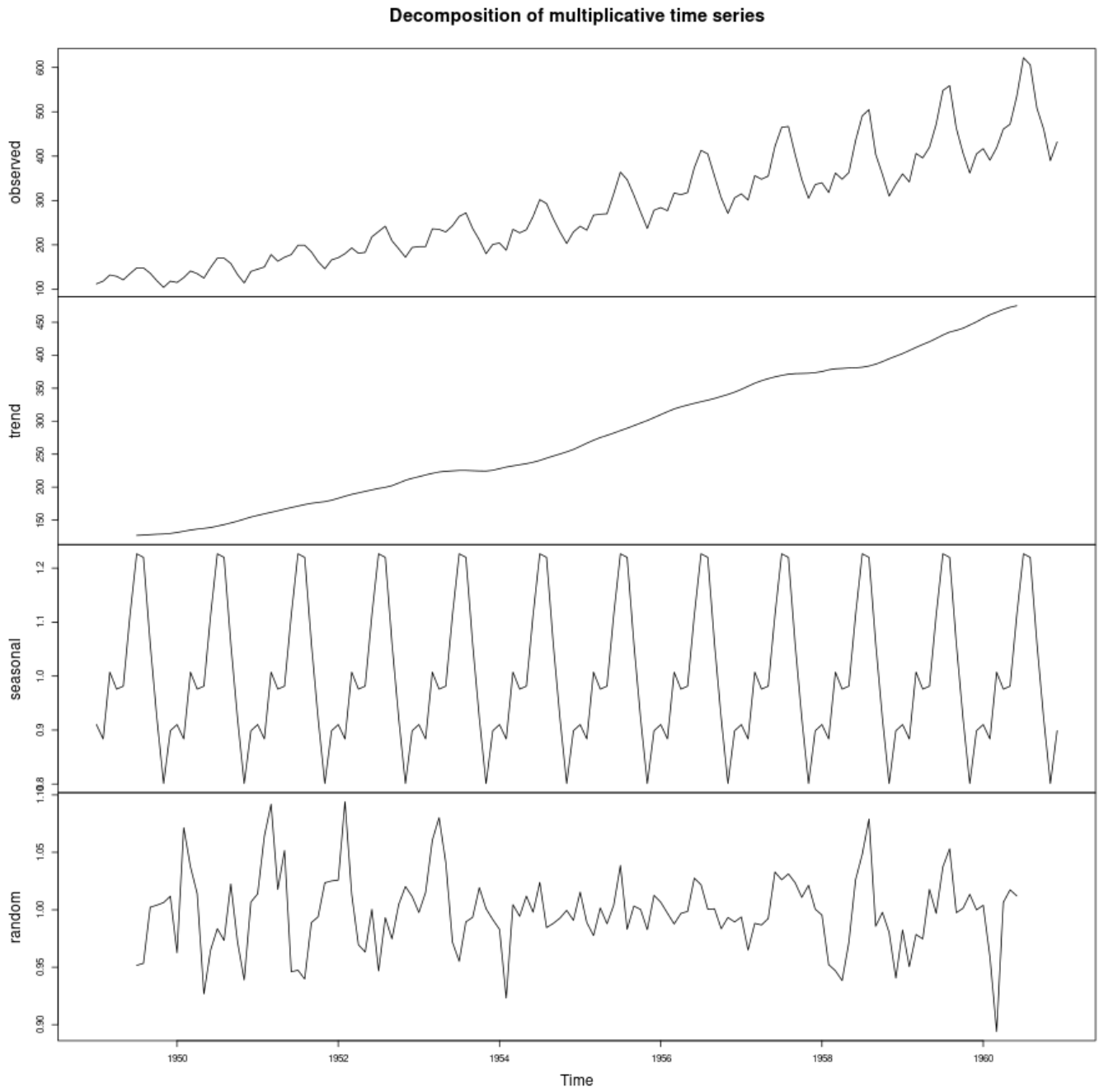
1. Data Overview

The dataset shows a strong upward trend and clear seasonality.



2. Decomposition

The decomposition confirms both an upward trend and a strong seasonal component.



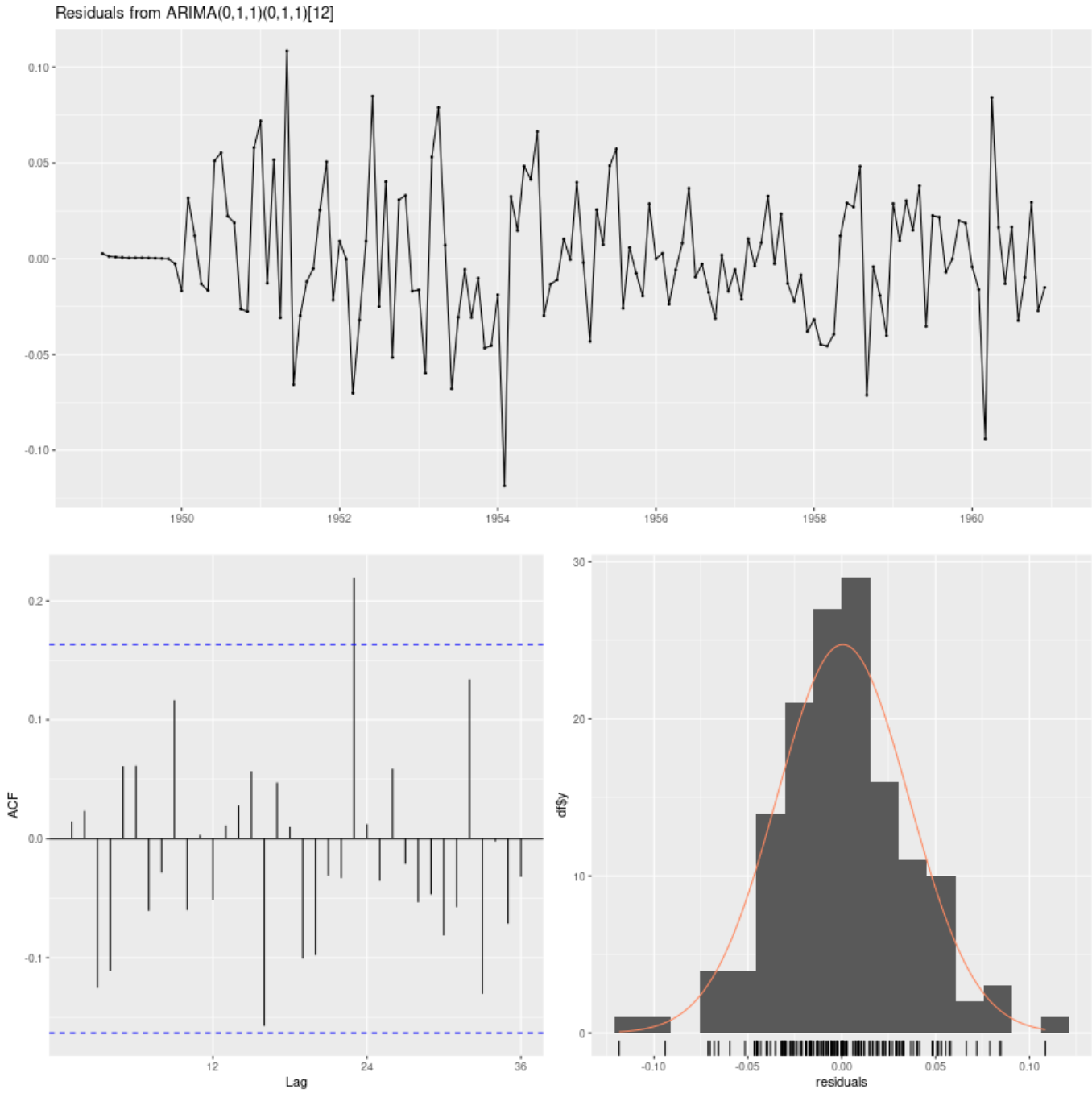
3. Model Identification

The KPSS test rejected stationarity ($p < 0.01$). The `auto.arima()` function selected a seasonal ARIMA model:

- **Model**: **SARIMA(0,1,1)(0,1,1)$_{12}$**
- **AIC**: -483.4
- **BIC**: -474.77

4. Goodness of Fit (Residual Diagnostics)

The **Ljung-Box** test gave a p-value of **0.233** (> 0.05), indicating that the residuals are randomly distributed (white noise). This confirms that the model is a good fit and can reliably be used for forecasting.

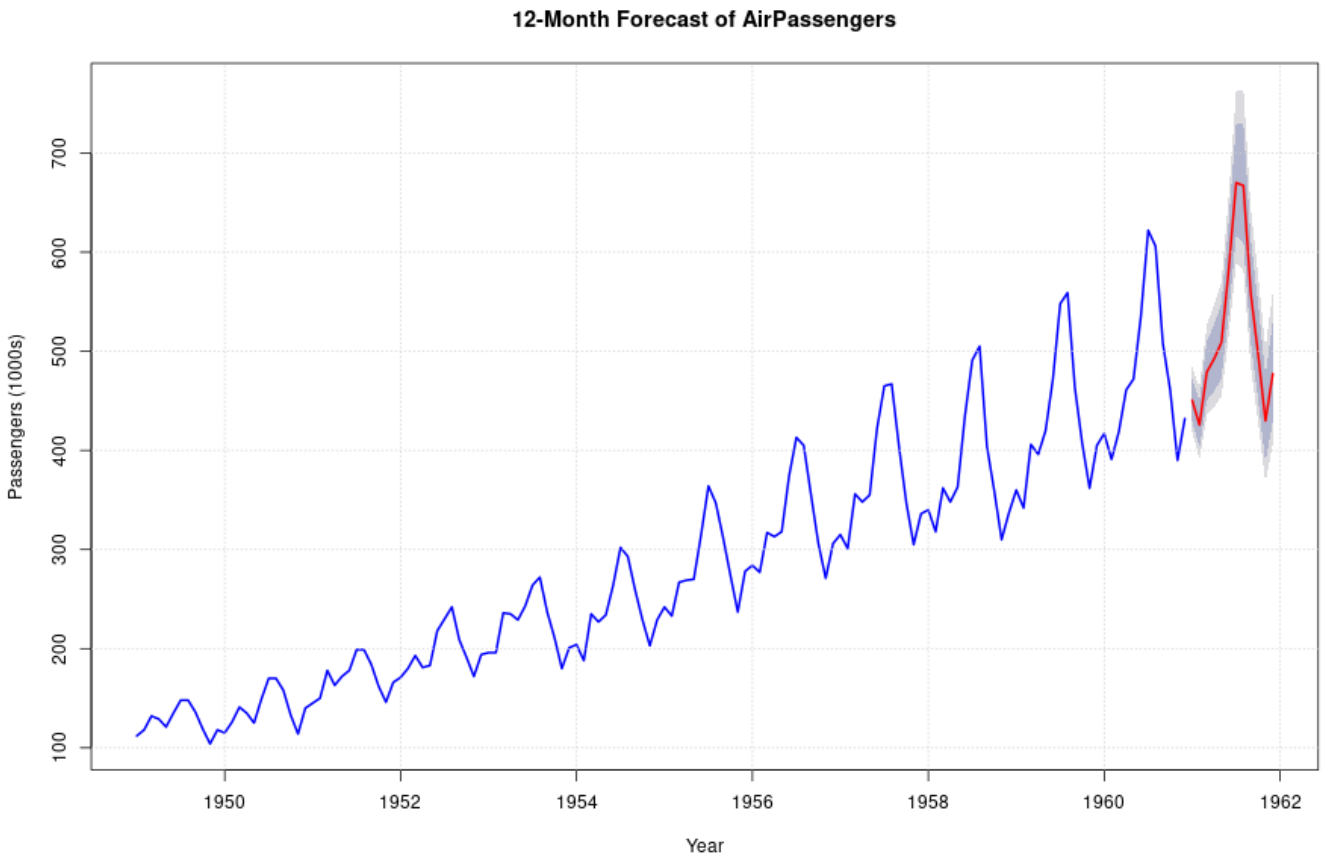


5. Forecast for 1961 (Next 12 Months)

The forecast continues both the growth trend and the seasonal patterns.

Month	Point Forecast (1000s)	95% Confidence Interval
Jan 1961	**450.4**	[418.9, 484.3]
Feb 1961	**425.7**	[391.2, 463.3]
Mar 1961	**479.0**	[435.6, 526.8]
Apr 1961	**492.4**	[443.5, 546.7]
May 1961	**509.1**	[454.6, 570.0]
Jun 1961	**583.3**	[516.8, 658.5]

Month	Point Forecast (1000s)	95% Confidence Interval
Jul 1961	**670.0**	[589.1, 762.1]
Aug 1961	**667.1**	[582.3, 764.2]
Sep 1961	**558.2**	[484.0, 643.8]
Oct 1961	**497.2**	[428.3, 577.2]
Nov 1961	**429.9**	[368.0, 502.1]
Dec 1961	**477.2**	[406.2, 560.7]



Conclusion

The forecasting model **SARIMA(0,1,1)(0,1,1)$_{12}$** predicts that airline passenger numbers will continue to grow through 1961, maintaining the established seasonal peaks (peaking in July/August) and troughs (trough in November).